



A BOUND ON PRICE IMPACT AND DISAGREEMENT

Philippe van der Beck, Lorenzo Bretscher, Julie Zhiyu Fu

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 - Elastic market view: No
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- **Why matters**: Can we understand asset prices from quantity data?
- **This paper**: Given observable moments (σ_p, σ_q) , If one:

Acknowledges

Investors disagree with each other



Must accept

Flows have large price impacts

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Formalize this tension in a statistical bound under fairly general assumptions:

$$\mathcal{M} \geq \frac{\sigma_p}{\sigma_q} \sqrt{\frac{1}{\rho} - 1}$$

THE PRICE IMPACT BOUND

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Connecting two observable moments:

- σ_p : return volatility
- σ_q : portfolio flow volatility ($\sim$ turnover)

to agreement ρ and price impact $\mathcal{M}$:

- $\rho \in [0, 1]$: Correlation in demand *shifts*
- $\mathcal{M}$: % Δp due to 1% demand shift

For average US stock: $\sigma_p \approx \sigma_q \approx 20\%$
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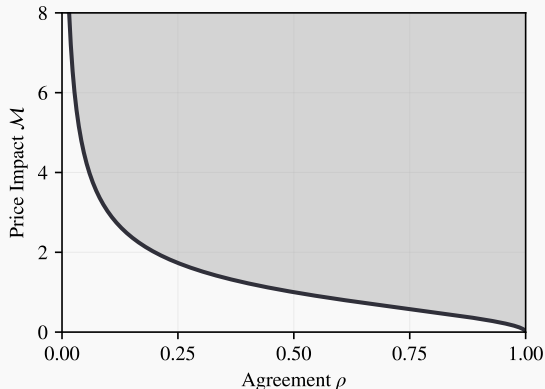
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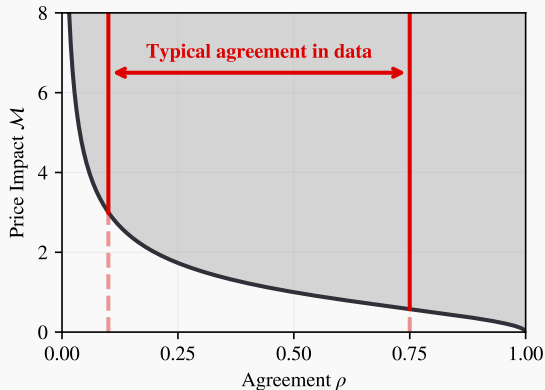
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- Log-linearization of generic portfolio choice problem

(Campbell and Viceira, 2002; Koijen and Yogo, 2019; Gabaix and Koijen, 2021)

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For practice: a simple & transparent metric for price impact

- A sanity check for estimates from event studies
- Ballpark estimates when event studies are hard to find

MODEL FRAMEWORK

A general framework: log-linear demand curve for a stock:

$$\Delta q_{i,t} = -\zeta_i \Delta p_t + u_{i,t}$$

where ζ_i is price elasticity and $u_{i,t}$ demand shifter.

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- Elasticity $\zeta^{CRRA} \equiv \frac{\overline{dp}}{dp - r_f}$: Expected return and risk-free rate
- Demand shifter u_i contains: changes in risk aversion γ_i , uncertainty, and (in dynamic setting) future expected price change $\mathbb{E}[\Delta p_{t+1}]$

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...depending on the microfoundation, ζ_i captures different structural parameters.

DERIVING THE BOUND WITH HOMOGENEOUS ELASTICITY

For illustrative purpose, start with homogeneous elasticity ζ

$$\left. \Delta q_{i,t} = -\zeta \Delta p_t + u_{i,t} \right\}$$

DERIVING THE BOUND WITH HOMOGENEOUS ELASTICITY

Let S_i denote ownership share, we have the market clearing condition:

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where price impact $\mathcal{M} := \frac{1}{\zeta}$, agg. (avg.) demand shifts $u_{S,t} := \sum_i S_i u_{i,t}$

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Key Object: The (size-weighted) average flow volatility $\sigma_q^2 := \sum_i S_i \sigma_{q,i}^2$

$$\sigma_q^2 = \left(\frac{1}{\rho} - 1\right) \times \text{Var}(u_{S,t}) \quad \rho := \frac{\text{Var}(u_{S,t})}{\sum_i S_i \text{Var}(u_{i,t})} \in [0, 1]$$

- Investor agreement ρ : R^2 of demand shifts explained by the cross-investor average (Loosely: How correlated are investors demand shifts?)

DERIVING THE BOUND WITH HOMOGENEOUS ELASTICITY

Putting everything together, we have:

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...When elasticities ζ_i are heterogeneous, the equality becomes a **lower bound**

TAKING THE BOUND TO THE DATA

For an asset n , estimate $\sigma_p(n)$ and $\sigma_q(n)$ from a 5-year backward-looking rolling window

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- 2 Estimate using **portfolio turnover**: sum of all position changes to share outstanding

$$\hat{\sigma}_{q,t}(n) \approx \hat{\mathbb{E}} \left[\frac{\sum_i |\Delta Q_{i,t}(n)|}{\bar{Q}(n)} \right]$$

Portfolio turnover is the $\mathcal{L}1$ approximation of the $\mathcal{L}2$ definition

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where $\Delta f_{i,t}(n)$ is the quarterly update for the 1q/2q/3q/LTG earnings forecast

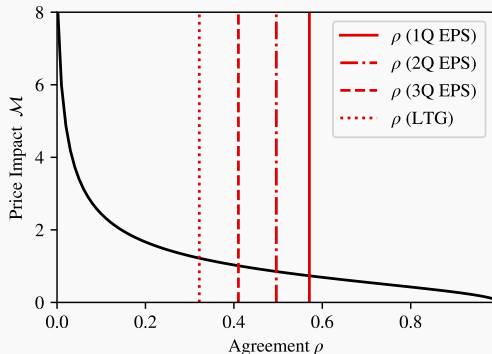
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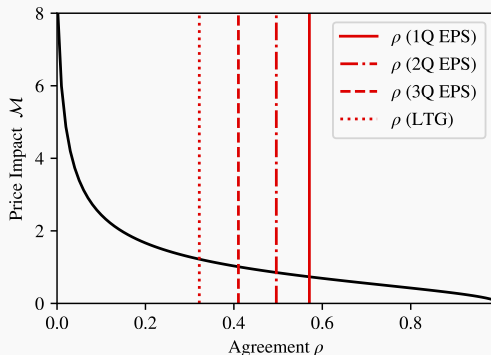
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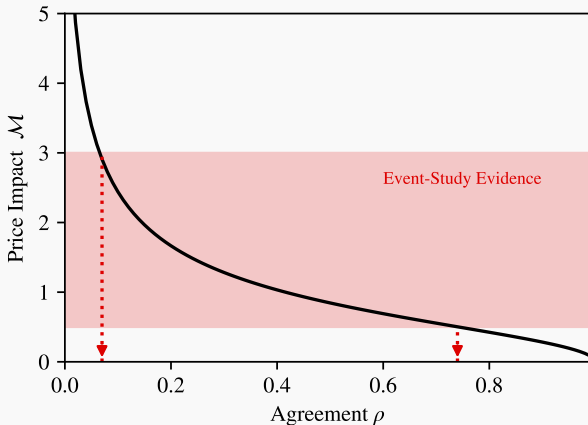
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Intentionally illustrative: Just one source of investor heterogeneity! [Details](#)

WHAT DO EVENT-STUDY ESTIMATES OF $\mathcal{M}$ IMPLY ABOUT ρ ?

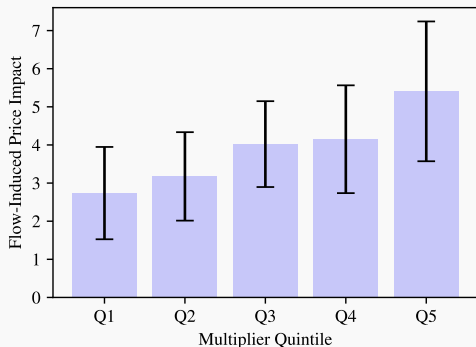
- Existing studies using event-study approach to estimate the price impact of an exogenous demand shift
- Our IBES estimates of ρ lie well within range implied by Event-Studies



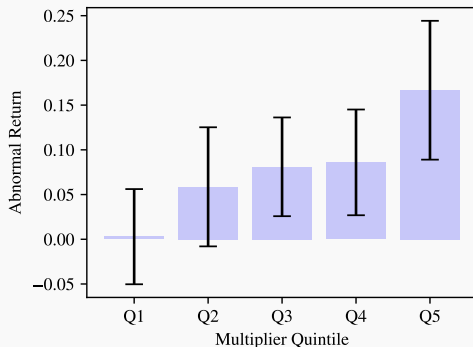
EMPIRICAL RELEVANCE OF THE BOUNDS

Our bounds explain cross-stock heterogeneity in event-based price impact estimates

Flow-Induced Trades (FIT):
estimates using mutual fund flows



S&P 500 Index Reconstitutions:
estimates using passive index demand



USING THE BOUNDS WHERE EVENT STUDIES ARE RARE

Use the simple p/q volatility ratio $\tilde{\mathcal{M}} = \frac{\sigma_p}{\sigma_q}$ to explore where event studies are rare

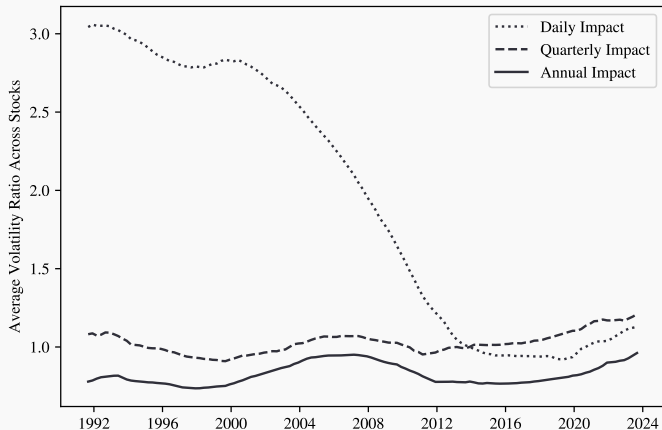
- Implicit assumption: ρ not too far away from 0.5 so $\sqrt{\frac{1}{\rho} - 1} \approx 1$

Exploring the heterogeneity across

- time & horizon “Rethinking Volume: Has the Market Become More Liquid?” (Beck, Bretscher, Fu)
- asset characteristics
- aggregation level of portfolios

PRICE IMPACT AT DIFFERENT HORIZONS AND ACROSS TIME

Construct p/q volatility ratio at different frequencies

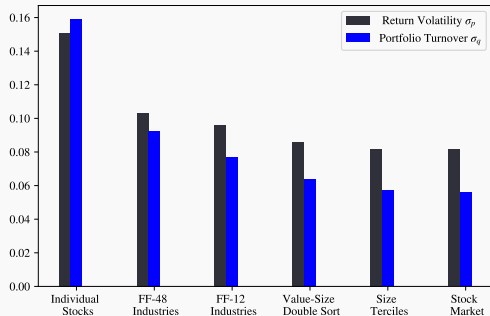


PRICE IMPACT AND THE CROSS-SECTION OF STOCKS

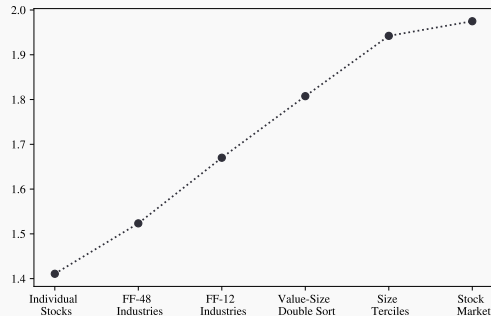
- Which assets are more inelastic?
- Regress price impact bound $\mathcal{M}$ onto stock-specific characteristics.
- Small, risky, momentum, growth and illiquid stocks command higher multipliers.

	$\mathcal{M}_{EPS}$			$\tilde{\mathcal{M}}$
	(1)	(2)	(3)	(4)
log(ME)	-0.396*** (0.017)	-0.383*** (0.017)	-0.675*** (0.033)	-0.593*** (0.029)
β	0.121*** (0.010)	0.159*** (0.011)	0.148*** (0.013)	0.134*** (0.012)
Cum. Ret.	0.160*** (0.013)	0.151*** (0.009)	0.138*** (0.007)	0.125*** (0.006)
BM	-0.112*** (0.013)	-0.111*** (0.012)	-0.127*** (0.012)	-0.111*** (0.011)
<u>Dividend</u> BE	0.015 (0.009)	0.023* (0.009)	-0.028* (0.011)	-0.026* (0.010)
Profit	-0.085*** (0.011)	-0.088*** (0.010)	-0.001 (0.011)	0.001 (0.010)
Amihud Illiquidity	0.261*** (0.017)	0.265*** (0.017)	0.182*** (0.014)	0.167*** (0.013)
Date FE	—	x	x	x
Stock FE	—	—	x	x

PRICE IMPACT AND PORTFOLIO AGGREGATION



- σ_p declines (diversification)
- σ_q declines (substitution).



- Price impact σ_p/σ_q becomes larger for more aggregated portfolios.

CONCLUSION

1 High σ_p and low σ_q implies a tension between disagreement and elasticity:

- If one acknowledges that investors disagree with each other
⇒ must accept that flows have large price impacts
- Formalized in the **price-impact bound**

$$\mathcal{M} \geq \frac{\sigma_p}{\sigma_q} \sqrt{\frac{1}{\rho} - 1}$$

- Empirically: correlation of demand shifts ρ does not seem excessively high **on average**

2 The p/q volatility ratio $\frac{\sigma_p}{\sigma_q}$: an easy-to-compute metric for price impact

- Sanity check for event-based estimates
- Ballpark estimates when event studies are hard to find

DYNAMICS: A SIMPLE EXAMPLE

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- Does this mean markets are super inelastic at announcement and very elastic at inclusion date? No. **Disagreement has changed over time.**
- Demand Curve: $\Delta q_{i,t} = -\zeta \Delta p_t + \underbrace{\mathbb{E}_t[\Delta p_{t+1}]}_{u_{i,t}} + d_{i,t}$
 - At announcement: $\mathbb{E}_t[\Delta p_{t+1}] = 5\%$ and $d_{i,t} = 0$ for both
Both have positive demand shift $u_{i,t} = 5\% \rightarrow$ **Agreement!**
 - At inclusion: $\mathbb{E}_{t+1}[\Delta p_{t+2}] = 0\%$ for both but different demand shifts.
 $u_{i,t} = -5\%$ for HF and $u_{i,t} = 5\%$ for BM. $\rightarrow$ **Disagreement!**

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LEARNING-FROM-PRICE MODELS

Learning-from-price models: $\frac{Q_i P}{W_i} = \frac{\mathbb{E}[D|\mathbf{s}_i, \mathbf{P}_t]/P - r}{\gamma_i \sigma^2}$

(Hellwig, 1980; Grossman and Stiglitz, 1980)

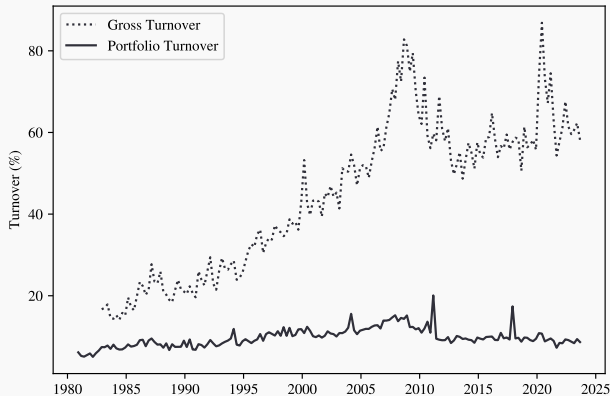
- Demand shifter $u_{i,t}$: private signal $\mathbf{s}_i$ for fundamental value
- Elasticity ζ_i : captures trade-off between noisy price vs. noisy private signal
- Information asymmetry drives heterogeneous portfolio choices
- Price informativeness affects both elasticity and demand shifts

PORTFOLIO TURNOVER VS TRADING VOLUME

Gross Turnover:

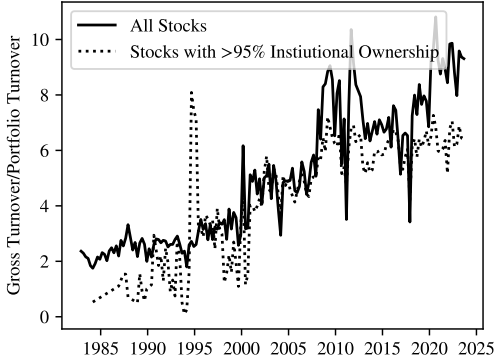
$$\frac{\text{Total Trading Volume in a quarter}}{\text{Shares Outstanding}}$$

- 8x larger than portfolio turnover!
- Inflated by round-trip trades
- We measure σ_p using *net* as opposed to *total* price changes. Should do same for volume!

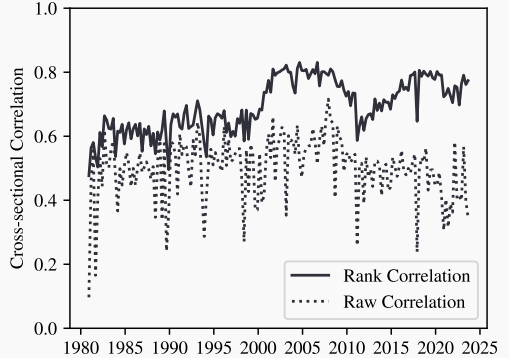


PORTFOLIO TURNOVER VERSUS GROSS TURNOVER

Gross Turnover/Portfolio Turnover

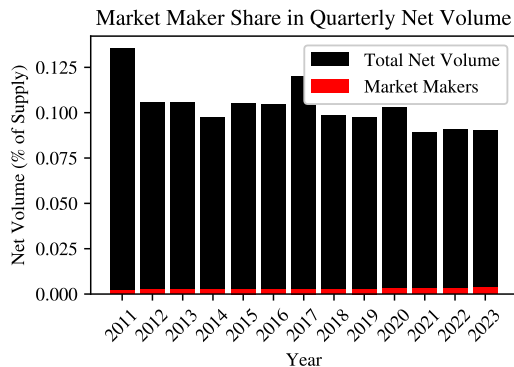


Cross-Sectional Correlation



PORTFOLIO TURNOVER VS TOTAL TRADING VOLUME

- Market making drives a lot of *total* trading volume
 - Citadel: about 25%
 - Jane Street: about 10%
 - SIG, Virtu, Two



→ However, their importance for institutional portfolio turnover is minimal!

THE BOUND IN A MULTI-ASSET SYSTEM

A common issue in estimating demand system is the cross-asset substitution

(Chaudhary et al., 2023; Haddad et al., 2025)

We show our bound still holds in multi-asset system, with more nuanced interpretation

- Consider such two-asset demand system:

$$\Delta \mathbf{q}_{i,t} = \Gamma \Delta \mathbf{p}_t + \mathbf{u}_{i,t}, \quad \Gamma = \begin{pmatrix} -\zeta(n) & \zeta(n, n') \\ \zeta(n', n) & -\zeta(n') \end{pmatrix} \quad (1)$$

Impose market clearing $\Delta \mathbf{q}_{S,t} = \mathbf{0}$ and solve the price for n' :

$$\Delta p_t(n') = \frac{\zeta(n', n)}{\zeta(n')} \Delta p_t(n) + \frac{1}{\zeta(n')} u_{S,t}(n').$$

SINGLE-PRICE REPRESENTATION OF DEMAND IN A MULTI-ASSET SYSTEM

Replace out the price of n' in the demand for asset n :

$$\Delta q_{i,t}(n) = - \underbrace{\zeta(n) (1 - Q_{n \leftarrow n'} Q_{n' \leftarrow n})}_{\tilde{\zeta}(n) = 1/\mathcal{M}(n)} \Delta p_t(n) + \underbrace{Q_{n \leftarrow n'} u_{S,t}(n') + u_{i,t}(n)}_{\tilde{u}_{i,t}(n)}, \quad (2)$$

where $Q_{n \leftarrow n'}$ and $Q_{n' \leftarrow n}$ are defined as

$$Q_{n \leftarrow n'} := \frac{\zeta(n, n')}{\zeta(n')}, \quad Q_{n' \leftarrow n} := \frac{\zeta(n', n)}{\zeta(n)}. \quad (3)$$

The bound still applies for the price impact $\mathcal{M}$ and the agreement on $\tilde{u}_{i,t}(n)$:

- Price impact $\mathcal{M}$ is no longer the inverse of the *own-price* elasticity $\zeta(n)$, but include the feedback effect from cross-asset substitution
- $\tilde{u}_{i,t}(n)$ includes the substitution effects coming from the demand shocks of other assets

EXAMPLE: SINGLE-STOCK ETF

- Consider two perfect substitutes: A stock and a single-stock ETF that tracks it.

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MORE RELAXED ASSUMPTION

Assumption

Let $\beta_i^u \equiv \frac{\text{Cov}(u_{i,t}, u_{S,t})}{\text{Var}(u_{S,t})}$ denote loading of investor i 's demand shift on the aggregate demand shift. We require:

$$\text{Cov}(\zeta_i, \beta_i^u) \leq \frac{1}{2} \mathcal{M} \text{Var}(\zeta_i)$$

where $\text{Var}(\cdot)$ and $\text{Cov}(\cdot)$ are the cross-sectional size-weighted sample moments.

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Intuition:

- More elastic investors do not load too strongly on the aggregate demand shift.
- In reality: Elastic market participants (liquidity providers) absorb, rather than amplify, common demand shifts: So $\text{Cov}(\zeta_i, \beta_i^u) < 0$.
- Effectively this rules out pathological case where investors who have more positive demand shifts than aggregate end up selling.

EXAMPLES: WHY $\text{COV}(\zeta_i, \beta_i^u)$ LIKELY NEGATIVE?

- Passive/benchmark-constrained investors, pensions, insurers, and target-date funds are price-insensitive by design (low ζ_i).
 - Demand shifts are driven by client flows and macro savings/liability cycles, which are common (high, positive β_i^u)
 - Low ζ_i , high β_i^u yields negative covariance.
- Liquidity providers: Dealers, APs/ETF arbitrage, HFT/market makers, and stat-arb funds are elastic (high ζ_i).
 - Their idiosyncratic shifts reflect inventory control or relative-value signals, typically orthogonal or contrarian to aggregate flows (low, negative β_i^u)
 - high ζ_i , low β_i^u yields negative covariance.
- Event microstructure: Around index inclusions/rebalances, the inelastic side (indexers, mandate followers) is the source of common shocks (high β_i^u), while the elastic side (dealers/APs/arbs) takes the other side.
- Funding and risk management: If a manager's shocks are strongly procyclical, being highly elastic would magnify drawdowns when liquidity is scarce.

AGGREGATION DETAILS

- Let g denote a level of aggregation (e.g., industry portfolios). Flows by investor i into group g are given by

$$\Delta D_{i,t}(g) = \sum_{n \in g} \Delta Q_{i,t}(n) P_{t-1}(n)$$

- Total trading volume relative to the portfolio's market cap $\bar{D}_{t-1}(g)$ is

$$V_t(g) = \frac{\sum_{i=1} |\Delta D_{i,t}(g)|}{\bar{D}_{t-1}(g)} \quad (4)$$

- This nets out roundtrip trades:

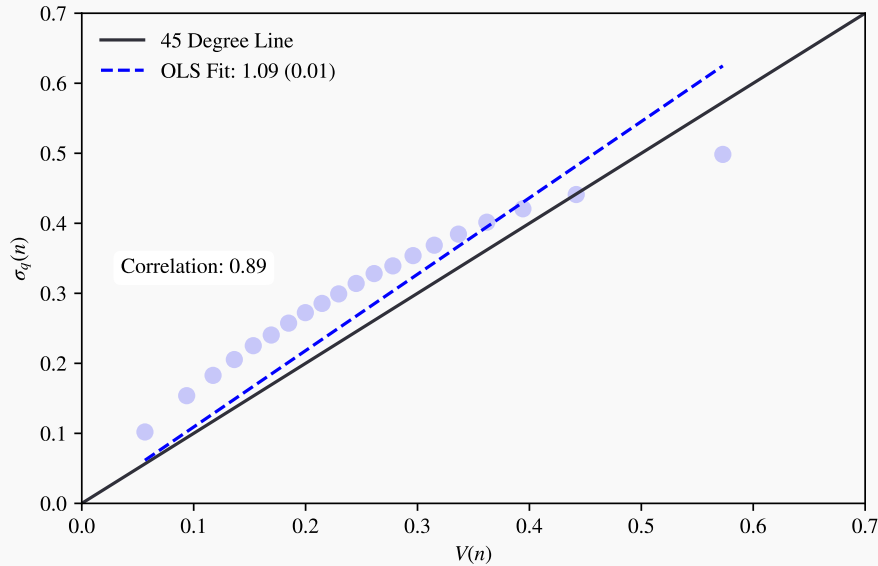
1

over time

2

across assets n within the portfolio A

$\mathcal{L}_1$ **VERSUS** $\mathcal{L}_2$



SUMMARY STATS IBES

Horizon	N Periods	N Forecasters	N Updates per Periods	N Updates
0Q	37.1	21.7	9.6	355.0
1Q	44.2	26.0	11.5	510.2
2Q	41.3	25.0	10.9	452.4
3Q	37.6	22.6	10.1	379.8
LTG	16.5	5.2	3.5	57.7

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FLOW-INDUCED TRADES (FIT)

- Regress returns onto flow-induced trades $FIT(n)$ and interaction $FIT(n) \times \mathcal{M}(n)$.

	Ret.		
	(1)	(2)	(3)
FIT	3.820*** (0.510)	2.635*** (0.636)	
$\mathcal{M}_{EPS}$		0.004 (0.004)	
$FIT \times \mathcal{M}_{EPS}$		1.256* (0.565)	
$FIT \times \mathcal{M}_{EPS}$ quintile: 1			2.755*** (0.624)
$FIT \times \mathcal{M}_{EPS}$ quintile: 2			3.175*** (0.588)
$FIT \times \mathcal{M}_{EPS}$ quintile: 3			4.029*** (0.577)
$FIT \times \mathcal{M}_{EPS}$ quintile: 4			4.152*** (0.718)
$FIT \times \mathcal{M}_{EPS}$ quintile: 5			5.485*** (0.938)
Date	x	x	x
$\mathcal{M}_{EPS}$ quintile	-	-	x
Observations	152862	152862	152862
R^2	0.249	0.250	0.250
R^2 Within	0.004	0.005	0.005

SP INCLUSIONS

- Regress abnormal SP inclusion returns onto $\mathcal{M}(n)$.

	Abnormal Return		
	(1)	(2)	(3)
$\mathcal{M}_{\text{EPS}}$		0.058* (0.025)	
$\mathcal{M}_{\text{EPS}}$ quintile: 1			0.046*** (0.014)
$\mathcal{M}_{\text{EPS}}$ quintile: 2			0.082*** (0.024)
$\mathcal{M}_{\text{EPS}}$ quintile: 3			0.073*** (0.016)
$\mathcal{M}_{\text{EPS}}$ quintile: 4			0.076*** (0.016)
$\mathcal{M}_{\text{EPS}}$ quintile: 5			0.140*** (0.024)
Intercept	0.080*** (0.008)	0.039 (0.021)	
Observations	837	686	686
R^2	0.021	0.036	0.040
Adj. R^2	0.017	0.029	0.027

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